FORD

Powered by Azure OpenAI + RAG

Recall Readiness Co-Pilot

An Agentic RAG-Based Warranty Risk Intelligence System for Proactive Recall Decision Support

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Platform: Azure OpenAI + Azure AI Search

Project Overview

Design Philosophy

As an AI Solution Designer, I apply a business-first approach to AI architecture. The system was designed to solve a real enterprise pain point: Ford's inability to detect recall-worthy failure patterns early enough to prevent reactive, costly recall events.

The design philosophy centers on three principles: contextual grounding, decision traceability, and responsible AI output — ensuring every insight the system generates is explainable, cited, and actionable.

What This System Does

- Ingests warranty claims, TSBs, and NHTSA data
- Reasons across multiple data sources agentically
- Outputs structured risk intelligence with confidence scores
- Supports proactive recall decisions before escalation
- Cites every source — no unsupported claims

Business Problem

Ford Motor Company — The Hidden Cost of Reactive Recall Management

$3.4B

Annual warranty cost (Ford 10-K)

18–24

Months: avg. time from signal to recall

50K+

NHTSA complaints received annually

The Core Gap

Ford's warranty engineering and quality teams receive fragmented signals — NHTSA complaint spikes, Technical Service Bulletin (TSB) patterns, early warranty claim clusters — from disconnected systems. No single intelligent layer synthesizes these signals into actionable recall readiness decisions in real time.

The result: engineers act reactively, recalls are delayed, and recall escalation costs Ford an estimated $100M+ per major reactive event.

- Siloed data across 5+ disconnected systems
- Manual analyst effort: 15+ hrs/week per engineer
- Reactive recalls avg. 3x more costly than proactive
- No confidence scoring on risk signals

Generative AI Use Case

Ford Recall Readiness Co-Pilot

Who Uses It

Warranty Engineers

Identify failure pattern clusters across model years

Quality & Safety Teams

Monitor NHTSA complaint spikes and TSB correlations

Legal & Compliance

Assess regulatory exposure and recall readiness risk

Three Core Decision Outputs

Should Ford issue a proactive recall?

Yes / No / Monitor with confidence score

Which components need redesign priority?

Ranked component risk list with signal evidence

What is projected warranty exposure risk?

High / Medium / Low with trend direction

Why RAG for Recall Intelligence?

Dimension	Traditional ML	Agentic RAG (This System)
Knowledge Source	Fixed training data only	Dynamic: docs + live NHTSA + structured DB
Explainability	Black-box predictions	Every response cites retrieved sources
Freshness	Stale until retrained	Real-time retrieval on every query
Recall Signal Coverage	Historical patterns only	Live complaint spikes + emerging TSB patterns
Decision Support	Scores without context	Structured output: risk + confidence + signals
Hallucination Risk	High without grounding	Mitigated via grounded generation constraint

"Recall decisions require continuously evolving external signals, making RAG essential for contextual accuracy and traceability."

Agentic RAG Architecture

Data Sources
Corpus + NHTSA + Claim DB

→

Chunk & Embed
text-embedding-ada-002
Azure OpenAI

→

Azure AI Search
Vector store
Semantic retrieval

→

Agentic Retriever
Multi-step reasoning
Context aggregation

→

GPT-4o Synthesis
Grounded generation
Hallucination mitigation

→

Risk Output
Score + Confidence+ Recommendation

Three-Layer Data Architecture

Layer 1: Static Corpus

TSBs, warranty policies, claim summaries — chunked, embedded, stored in Azure AI Search vector index

Layer 2: Live NHTSA API

Real-time complaint filings, recall announcements — retrieved dynamically on each query via web search tool

Layer 3: Structured Claim DB

Simulated Ford claim database (CSV/SQLite) queried via structured retrieval — hybrid RAG approach

How the Agent Reasons

Sample Query: *"What failure patterns in the 2022 F-150 power steering system correlate with recent NHTSA complaints, and what is the recall readiness risk?"*

01	**Query Decomposition**	Agent breaks query into sub-tasks: search TSB corpus for steering failures, retrieve NHTSA complaints, query claim DB for 2022 F-150 patterns
02	**Multi-Source Retrieval**	Semantic search across vector store retrieves top-k relevant TSB chunks. Parallel web retrieval pulls live NHTSA complaint filings. Structured query extracts claim counts from DB
03	**Context Aggregation**	Retrieved chunks from all three sources are merged into a unified context window. Relevance scoring filters low-confidence chunks before synthesis
04	**Relevance Ranking**	Agent scores each retrieved passage by semantic similarity and recency. Only passages above confidence threshold are passed to the generation layer
05	**Response Synthesis**	GPT-4o synthesizes grounded responses using only the retrieved context. Grounded Generation Constraint prevents any output not supported by the retrieved evidence

Decision Output Structure

The Ford Recall Readiness Co-Pilot returns a structured intelligence report on every query

● FORD RECALL READINESS CO-PILOT | OUTPUT REPORT

RECALL READINESS RISK: **HIGH** ▲ INCREASING

CONFIDENCE SCORE: **84%**

KEY SIGNALS DETECTED:

→ NHTSA steering complaints +38% in last 60 days

→ Similar failure pattern: 2019 F-150 recall (Case #2019-0442)

→ Warranty claims trending upward: +22% MoM for 2022 MY

→ TSB #22-0187 issued but incident rate not declining

RECOMMENDATION:

Initiate engineering investigation for 2022 F-150 steering system. Evaluate proactive recall readiness.

Risk Level

HIGH / MEDIUM / LOW
based on signal severity and pattern match

Confidence Score

Percentage certainty based on retrieved evidence strength and source consistency

Trend Direction

Increasing / Stable / Decreasing — tracks signal trajectory over 30/60/90 days windows

Key Signals

Evidence retrieved from all three data sources with source citations

Recommendation

Specific, actionable decision output — no vague suggestions, always grounded

Data Sources & Management

Raw Sources ▶ Preprocessing ▶ Vector Store ▶ Retrieval ▶ Output

Layer 1: Static Document Corpus

Source: Uploaded synthetic documents (5 files)

Preprocessing: Chunked into 512-token segments, overlap 50 tokens, cleaned for formatting artifacts

Quality: High fidelity, domain-specific, validated for accuracy

Storage: Persistent vector index in Azure AI Search

Layer 2: Live NHTSA Web Retrieval

Source: NHTSA Complaints API + web search (real-time)

Preprocessing: Filtered by model year and component, structured JSON parsed to text, timestamped

Quality: Government-published, authoritative, updated daily

Storage: Temporary session cache — retrieved fresh on each query

Layer 3: Structured Claim Database

Source: Simulated Ford claim records (CSV/SQLite)

Preprocessing: Normalized by component code, model year, and claim date. null values imputed

Quality: Synthetic but structurally accurate - mirrors real claim schema

Storage: Local structured store, queried via SQL-augmented retrieval

Product Integration

Fully embedded within Ford's Azure ecosystem — no new infrastructure required

Warranty Engineering Portal

Azure Bot Service
Embedded chat widget

Ford Quality Dashboard

Power BI integration
Risk score tiles

Azure OpenAI Recall Co-Pilot Core Engine

ERP / Warranty Management System

REST API feed
Claim data sync

Mobile Alert Interface

Teams / mobile app
High-risk notifications

All components deployed within Ford's existing Microsoft Azure tenant — leveraging Azure Active Directory for access control, Azure Monitor for observability, and Azure Key Vault for secrets management

Value Proposition & ROI

40%

Reduction in recall decision time

From avg. 18 months to ~11 months

3–6

Months earlier issue detection

Before NHTSA complaint thresholds trigger

$100M+

Recall escalation costs prevented

Per major proactive vs. reactive recall event

15 hrs

Analyst time saved per engineer/week

Redirected to validation and strategy work

Benefit Category Breakdown

Benefit Category	Current State	With Co-Pilot
Signal Detection Speed	18–24 months reactive	3–6 months proactive
Analyst Research Time	15+ hrs/week manual aggregation	< 2 hrs with AI-assisted synthesis
Recall Cost Profile	$200M–$500M reactive recall avg.	$50M–$150M proactive recall avg.
Decision Confidence	Qualitative judgment, inconsistent	Quantified score with source citations
Regulatory Risk	Reactive NHTSA investigation exposure	Proactive engagement, reduced exposure

Validation & Accuracy Methods

Cross-Verification

- Internal warranty claim trends cross-checked against NHTSA complaint data
- Responses validated against historical recall patterns for same component families
- Mismatches flagged and escalated for human analyst review

Grounded Generation Constraint

- Model responses restricted strictly to retrieved context — no external knowledge injection
- If retrieved context is insufficient, system returns: 'Insufficient evidence — manual review required'
- This prevents hallucination at the architecture level, not just the prompt level

Guardrails & Source Citations

- Every response includes source citations: document name, NHTSA case ID, or DB record reference
- Out-of-scope queries (HR, finance, non-warranty topics) are blocked by system prompt guardrails
- Confidence scores below 60% threshold trigger a 'Low Confidence - Verify Manually' flag

Human-in-the-Loop Feedback

- Warranty engineers validate or override every High-risk recommendation
- Validated outcomes fed back to refine retrieval relevance scoring over time
- Monthly accuracy audits compare Co-Pilot recommendations against actual recall decisions

Live Demo — Before & After

BEFORE | Without Co-Pilot

- Analyst opens 5 separate systems: warranty DB, NHTSA portal, TSB repository, claim analytics tool, engineering notes
- Manual search across each system: 2-3 hours per investigation
- Results compiled into spreadsheet — no consistency, no confidence scoring
- Risk assessment based on qualitative judgment — varies by analyst experience
- Decision submitted to committee — additional 2-4 weeks review cycle
- Total time from signal to decision: weeks to months

VS

AFTER | With Co-Pilot

- Engineer types one natural language query into the Co-Pilot interface
- Agent autonomously queries all three data sources in parallel — seconds
- Structured output returned: Risk Level, Confidence Score, Key Signals
- Every signal cited with source — NHTSA case IDs, TSB numbers, claim records
- Recommendation delivered with confidence score — ready for committee review
- Total time from signal to structured insight: under 60 seconds

Summary & Conclusion

Business Problem Solved

Ford's reactive recall process costs $100M+ per event. The Co-Pilot enables proactive, data-grounded decision support.

Technical Innovation

Agentic RAG with three-layer retrieval, grounded generation constraint, and confidence-scored structured output.

Responsible AI Design

Human-in-the-loop validation, source citations on every response, and guardrails preventing unsupported claims.

Enterprise Readiness

Azure-native deployment, existing tenant integration, Power BI and ERP connectivity — zero new infrastructure.

AI Solution Design Philosophy

- ✓ Business problem first, technology second
- ✓ Every AI output must be explainable and cited
- ✓ Confidence scoring enables trust, not just answers
- ✓ Human oversight is a feature, not a limitation
- ✓ Enterprise AI must be deployable on existing infrastructure

Future Scope & Extensions

Phase 2 roadmap for evolving the Co-Pilot into a full Recall Intelligence Platform

Phase 2

IoT Vehicle Telemetry Integration

Ingest real-time sensor data from connected Ford vehicles via Azure IoT Hub. Detect failure precursors before field complaints emerge, shifting from reactive claims analysis to true predictive recall intelligence.

Phase 3

Predictive Recall Modeling

Layer a statistical warranty maturation model beneath the RAG system. Forecast claim volume trajectory by component and model year — combining generative AI explainability with quantitative exposure prediction.

Phase 4

Cross-OEM Benchmarking

Expand retrieval corpus to include public recall filings across GM, Stellantis, and Toyota. Enable Ford engineers to identify industry-wide component vulnerability patterns and benchmark recall response speed against competitors.